

MOHAMMED RIZWAN N MASUTI

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PROFESSIONAL SUMMARY

Results-driven Aeronautical Engineering graduate with practical experience in aerospace systems, CFD simulations, propulsion analysis, and aircraft structural engineering through internships at Hindustan Aeronautics Limited (HAL) and S&I; Engineering Solutions. Skilled in ANSYS Fluent, HiFUN, CATIA, and 3DEXPERIENCE with strong expertise in compressible flow analysis, aerodynamic optimization, shockwave behavior, and propulsion system performance evaluation. Demonstrated ability to solve engineering problems through simulation-driven analysis and technical research. Passionate about aerospace innovation, aircraft development, and advanced propulsion technologies with strong adaptability to international engineering environments.

EDUCATION

- Bachelor of Engineering (B.E.) – Aeronautical Engineering | Global Academy of Technology, Bengaluru | 2021 – 2025

Relevant Coursework: Aerodynamics, Aircraft Structures, Propulsion Systems, Flight Mechanics, CFD, Aerospace Materials, Avionics

TECHNICAL SKILLS

- ANSYS Fluent, HiFUN, CATIA, 3DEXPERIENCE
- Basic Python, MATLAB, C++
- Computational Fluid Dynamics (CFD)
- Aerodynamics & Compressible Flow
- Aircraft Structures & Materials
- Propulsion Systems & Shockwave Analysis
- Technical Documentation & Engineering Data Analysis

INTERNSHIP EXPERIENCE

Engineering Intern – Hindustan Aeronautics Limited (HAL), Bengaluru | Nov 2024 – Dec 2024

- Studied aircraft structural systems including fuselage, wing assemblies, empennage, and landing gear mechanisms.
- Observed aerospace manufacturing, assembly, and aircraft maintenance operations in real-time engineering environments.
- Learned aerospace-grade material selection techniques focused on structural integrity and weight reduction.
- Gained practical exposure to propulsion systems, aerodynamics, and aircraft operational mechanisms.
- Improved understanding of aerospace workflows, engineering standards, and industrial safety procedures.

CFD Intern – S&I; Engineering Solutions Pvt. Ltd., Bengaluru | Apr 2025 – May 2025

- Conducted CFD simulations for ramjet inlet flow analysis using aerospikes geometry under compressible flow

conditions.

- Utilized ANSYS Fluent and HiFUN to evaluate shockwave structures, Mach number distribution, and pressure recovery.
- Optimized inlet geometries to improve propulsion efficiency and reduce aerodynamic losses.
- Compared solver convergence, computational efficiency, and simulation accuracy for engineering validation.
- Interpreted post-processing results and prepared technical performance reports with design improvement recommendations.

ACADEMIC PROJECT

Design and Analysis of Flow Characteristics for Ramjet Inlet using Aerospike

- Designed multiple ramjet inlet configurations for high-speed aerospace propulsion applications.
- Performed advanced CFD simulations to study compressible airflow behavior and aerodynamic efficiency.
- Investigated shockwave interactions, pressure recovery, and flow separation characteristics.
- Validated simulation outcomes with reference experimental data to improve reliability.
- Optimized aerodynamic geometry for enhanced airflow management and propulsion performance.

CERTIFICATIONS

- Skill Development Training on Drone Technology – Pegasus Aerospace
- Mastering Composites: From Fundamentals to Fabrication – Decadrives
- Fundamentals of Avionics – Global Academy of Technology
- GCAAMS 2024 – Participation

LANGUAGES

- English – Professional Working Proficiency
- German – Basic Understanding (Beginner Level, Non-Certified)